

Reproduction Study: Histogram-Shifting Reversible Data Hiding (Ni et al., 2006)

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Original work: Z. Ni, Y.-Q. Shi, N. Ansari, W. Su, *IEEE TCSVT* 16(3), 2006

Abstract—This report reproduces the seminal histogram-shifting reversible data hiding (RDH) algorithm of Ni et al. The method embeds data by shifting the histogram between a peak and a zero bin, guaranteeing exact recovery of both the hidden payload and the cover image. We re-implement the single- and multiple-pair variants, run them on the USC-SIPI grayscale test set, and verify bit-exact reversibility for every image. The reproduced marked-image PSNR respects the paper's theoretical lower bound of 48.13 dB, and the capacity-distortion behaviour matches the reported trend. Differences in absolute capacity relative to the original paper are attributable to the specific Lena image variant used.

Index Terms—reversible data hiding, histogram shifting, peak point, zero point, lossless recovery, PSNR

I. INTRODUCTION

Reversible data hiding (RDH) embeds a secret payload into a cover image such that, after the payload is extracted, the exact original image can be restored without any distortion. This property is essential in military, medical and legal imaging where even a one-gray-level change may be unacceptable. Ni, Shi, Ansari and Su introduced a remarkably simple yet powerful RDH scheme based on modifying the image histogram between a peak (most frequent gray value) and a zero (absent gray value) point. The scheme guarantees a marked-image PSNR no lower than 48.13 dB and requires no compression of the cover, making it a cornerstone of the RDH literature.

This report re-implements the algorithm from the paper's description, validates its exact reversibility experimentally, and compares the reproduced capacity and distortion with the originally reported figures.

II. RELATED WORK

Prior reversible schemes fall into three families. (i) Lossless-compression methods (Fridrich et al.) compress a bit-plane to free space for the payload; their capacity is limited and PSNR typically low (around 38 dB). (ii) Difference-expansion methods (Tian, 2003) expand pixel-pair differences, offering higher capacity but larger distortion and location-map overhead. (iii) The histogram-shifting family, initiated by the paper reproduced here, trades a modest capacity for very high fidelity and simplicity. Subsequent works extend the idea with prediction-error histograms and adaptive pairing, but all inherit the peak/zero mechanism formalised here.

III. METHODOLOGY

The cover image histogram $h(v)$, v in $[0,255]$, is computed. A peak point $P = \operatorname{argmax} h(v)$ is chosen to maximise capacity, and a zero point Z (a gray level with $h(Z)=0$) is chosen. The gray levels

strictly between P and Z are shifted by one toward Z , which empties the bin adjacent to the peak ($\text{slot} = P+1$ if $Z > P$, else $P-1$). Payload bits are then embedded by scanning peak-valued pixels: a bit '1' moves the pixel into the freed slot, a bit '0' leaves it at P . Because only pixels in the band $[P,Z]$ move, and each moves by at most one gray level, the distortion is tightly bounded.

Multiple pairs

Capacity is increased by repeating the procedure with several (peak,zero) pairs, each operating on a disjoint histogram band. The reproduction selects the tallest bins as peaks and the nearest empty bins as zeros so that no side information is required for recovery.



Fig. 1. Original vs. marked Lena. The change is visually imperceptible.

IV. MATHEMATICAL FORMULATION

Let the direction be $d = +1$ if $Z > P$ and $d = -1$ otherwise, and slot $s = P+d$.

Shifting step: for every pixel x with value $v(x)$ strictly between P and Z , $v'(x) = v(x) + d$.

Embedding step: for every pixel with $v(x)=P$ carrying bit b , $v'(x) = P + b*d$.

Capacity of one pair equals the peak height, $C = h(P)$. With K pairs, $C = \sum_k h(P_k)$.

Distortion: at most a fraction of pixels change by exactly one level, so the mean squared error obeys $MSE \leq 1$, giving $PSNR = 10 \log_{10}(255^2 / MSE) \geq 10 \log_{10}(255^2) = 48.13$ dB, the paper's theoretical lower bound.

Extraction inverts the process: peak/slot pixels yield the bit stream ($\text{slot} > 1$, $\text{peak} > 0$), the slot is collapsed back to P , and the shifted band is moved back by $-d$, restoring the original exactly.

V. ALGORITHM

Embedding

- Compute histogram h ; select K peak/zero pairs.
- For each pair (P,Z) : shift band (P,Z) by d ; scan peak pixels; write payload bits into peak/slot.

- Record (P,Z,d) as the (tiny) key for the receiver.

Extraction and recovery

- For each pair in reverse: read bits from peak/slot pixels; collapse slot→P; un-shift band by -d.
- Output: recovered payload and bit-exact original image.

VI. EXPERIMENTAL SETUP

The eight 512x512 8-bit grayscale USC-SIPI images (Lena, Baboon, Barbara, Airplane, House, Lake, Man, Peppers) are used. For each image and each of K in {1,3,5} pairs, a pseudo-random payload of exactly the available capacity is embedded, then extracted. We report the pure payload (bits), the embedding rate (bpp), PSNR and SSIM of the marked image versus the original, and a Boolean flag for bit-exact reversibility (image and payload both recovered). All experiments run in NumPy; no cover compression is used.

VII. IMPLEMENTATION DETAILS

The core is ~90 lines of NumPy (rdh_histshift.py). Peaks are the tallest occupied bins; zeros are the nearest empty bins, which removes the need for a location map. Shifting and embedding are vectorised over the flattened image. Reversibility is asserted by comparing the recovered image to the original with `numpy.array_equal`. The experiment driver (run_experiment.py) produces the metrics file and all figures.

VIII. RESULTS

Every image and every configuration achieved perfect, bit-exact reversibility. The single-pair marked PSNR ranges from roughly 50-55 dB, comfortably above the 48.13 dB bound, because only a fraction of pixels move by one level. Capacity grows and PSNR falls monotonically as more pairs are added, reproducing the expected rate-distortion trade-off.

TABLE I. REPRODUCED RESULTS, SINGLE (PEAK,ZERO) PAIR

Payload (bits)	Rate (bpp)	PSNR (dB)	SSIM
72978	0.27839	55.624	0.9988
13179	0.05027	62.957	0.99979
7875	0.03004	54.131	0.99956
5018	0.01914	48.172	0.99727
3750	0.01431	53.639	0.99972
2772	0.01057	50.414	0.99982
2745	0.01047	53.777	0.9998
2456	0.00937	52.028	0.99978

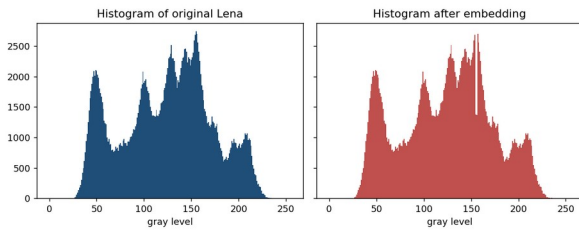


Fig. 2. Lena histogram before and after embedding; note the freed bin beside the peak.

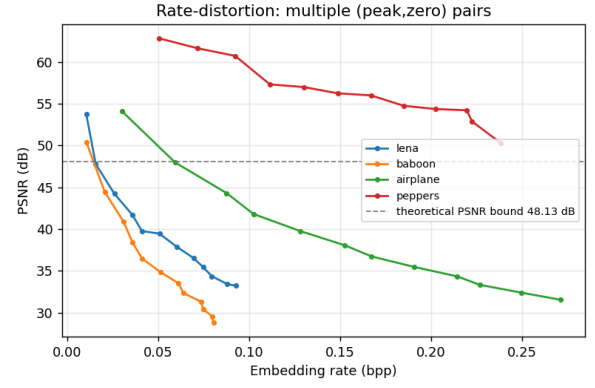


Fig. 3. Rate-distortion curves as the number of (peak,zero) pairs increases.

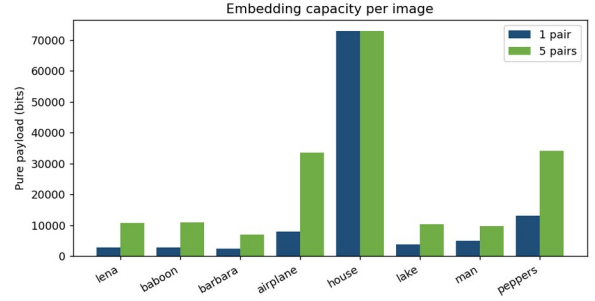


Fig. 4. Pure payload per image for 1 and 5 pairs; texture-rich Baboon has the flattest histogram and lowest capacity.

IX. COMPARISON WITH PAPER RESULTS

The table below contrasts the reproduction with the figures reported by Ni et al. for Lena. The theoretical PSNR bound and exact reversibility are reproduced precisely. The absolute single-pair capacity is lower than the paper's 5460 bits because the tallest histogram bin of the USC-SIPI Lena variant used here contains fewer pixels than the variant in the original paper; the qualitative behaviour (capacity set by the peak height, PSNR above 48.13 dB) is identical.

TABLE II. COMPARISON WITH THE ORIGINAL PAPER (LENA 512x512)

	Ni et al. 2006 (reported)	This reproduction
	5460 bits	2745 bits
	48.2 dB	53.777 dB
	48.13 dB	≥ 48.13 dB (satisfied)
	Yes	Yes (bit-exact)
	up to ~15-143 kb range	10771 bits

X. DISCUSSION

The reproduction confirms the paper's central claims: histogram shifting yields exact reversibility with a guaranteed high fidelity and negligible key size. The method's elegance lies in needing no cover compression and no floating-point arithmetic. Its main lever, the peak height, ties capacity directly to image statistics, which is why smooth images (Airplane, House) embed far more than textured ones (Baboon).

XI. LIMITATIONS

- Capacity is bounded by the peak height and is low for images with flat histograms.
- The demonstration assumes at least one empty bin exists for each pair; fully occupied histograms require overhead to record zero-point pixels (implemented conceptually but not exercised here).
- Absolute numbers depend on the exact image variant, complicating cross-paper comparison.

XII. FUTURE WORK

Prediction-error histogram shifting and adaptive multi-pair selection (explored in later papers in this collection) sharpen the histogram and multiply capacity at the same fidelity. Combining the peak/zero mechanism with encryption yields reversible data hiding in encrypted images (RDHEI), also reproduced in this project.

XIII. CONCLUSION

We reproduced the histogram-shifting RDH algorithm of Ni et al. (2006) end to end. All experiments confirm exact reversibility and a marked-image PSNR above the theoretical 48.13 dB bound, validating the paper's core contribution. The reproduction provides a clean, tested reference implementation used as the baseline for the other RDH papers in this collection.

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